

Lipid peroxidation assessment

Lipid peroxidation (LP) is determined by quantifying the malondialdehyde (MDA) content in the homogenate supernatant in the caudal adjacent segment to epicenter by colorimetric reaction with thiobarbituric acid (TBA) at high temperatures. Malondialdehyde is the principal and most studied product of polyunsaturated fatty acid peroxidation.

Material:

1. Spinal cord segment
2. Micro tubes
3. TRIS-HCl buffer solution (10 mM, pH 7.4)
4. TBA 0.8 %
5. Acetic acid buffer
6. Distillated water
7. Sodium dodecyl sulfate 8.1 %

Procedures:

1. Collect the spinal cord fragment.
2. Cool immediately the fragment to -20 °C.
3. Homogenize the sample in a sonicator (Branson Sonifier Model 250; three 4-s cycles) in TRIS-HCl buffer solution (10 mM, pH 7.4), and centrifuge the homogenate at 2500 rpm and 4 °C for 10 min.
4. Initially, pipette 134 µL of TBA 0.8 %, 134 µL of acetic acid buffer (125 mL of H₂O, 18.84 mL of glacial acetic acid, and 5.63 mL of HCl PA 12 N; pH 3.4), and 54 µL of H₂O.
5. After incubation, add 54 µL of the sample and 54 µL of sodium dodecyl sulfate (SDS) 8.1 %, and leave the solution at 95 °C for 90 min.
6. The MDA content is measured at a wavelength of 532 nm.
7. The results were normalized by protein content.